Basic Principles of Signal Timing and Design



Review of signal timing terminologies

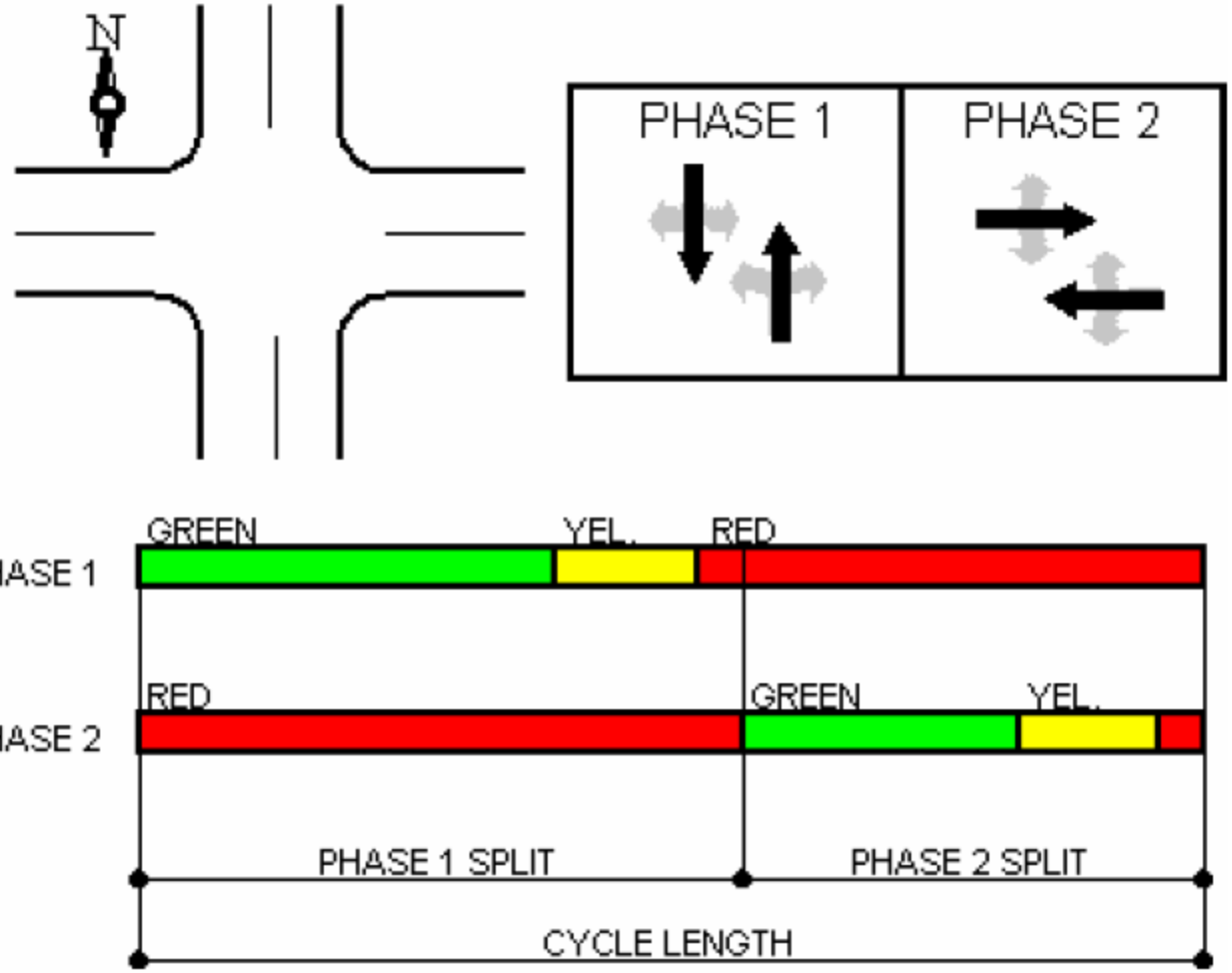
- **Cycle**: one complete rotation through all of the indications provided.
- **Phase**: a green interval, plus the change and clearance intervals that follow it.
- **Cycle length**, "C": time to complete one full cycle.
- **Green interval**, "G_i":
- **Yellow or change interval**, "Y_i"
- **All-red or clearance interval**, "AR_i"
- **Red interval** (sum of green, yellow, and all-red intervals for all other phases).



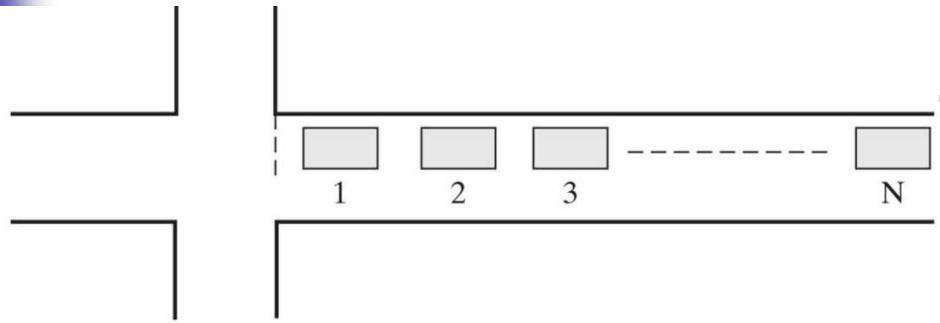




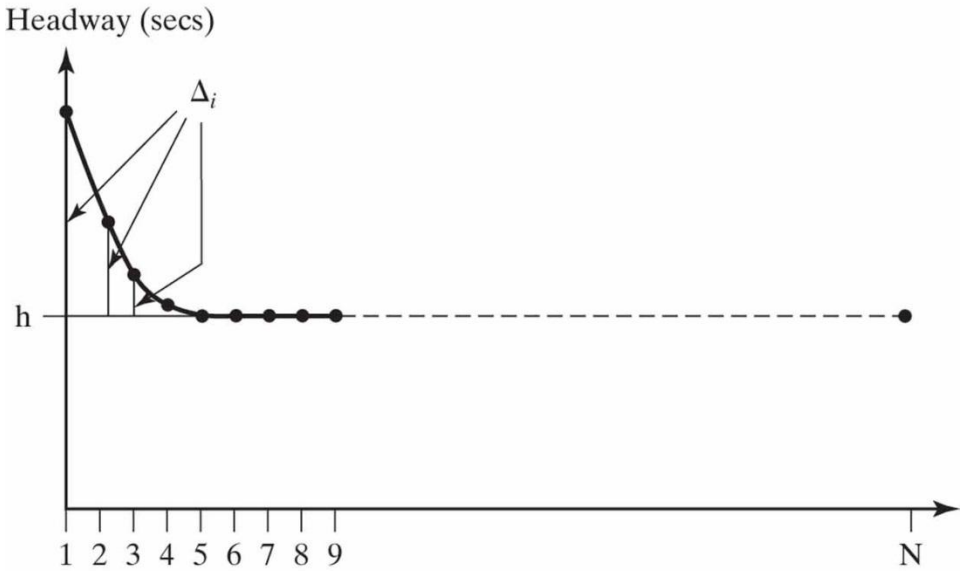
Illustration of a simple two-phase cycle



Saturation flow rate



(a) Vehicles in an Intersection Queue



(b) Average Headways Departing Signal

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$$s = \frac{3600}{h}$$

where:

s = saturation flow rate in veh/h,

h = saturation headway in s/veh

Start-up lost time

- **Start-up lost times** occur when a signal indication first turns from red to green, drivers in the queue do not instantly start moving at the saturation flow rate.

$$l_1 = \sum_i \Delta_i$$

- where:
- l_1 = start-up lost time, s/phase
- Δ_i = incremental headway (above "h" seconds) for vehicle i

Start-up lost time

- $T_n = l_1 + nh$

- T_n: green time required to move queue of n vehicles through a signalized intersection, s,
- L_1: start-up los time, s/phase
- n: number of vehicles in queue
- h: saturation headway, s/veh

Clearance lost time

- **Clearance lost time** occurs when the signal indication turns from green to yellow, the latter portion of time during the yellow interval is generally not utilized by traffic. Also, if there is an all-red interval, this time period is generally not utilized by traffic.

$$l_2 = Y + AR - e$$

- where:
- l_2 = clearance lost time, s
- Y = length of yellow interval, s
- AR = length of all-red interval, s
- e = encroachment of vehicles into yellow and all-red, s (default value of 2.0 seconds is used).

Total lost time

- Start-up and clearance lost times are summed to arrive at a **total lost time** for the phase, given as

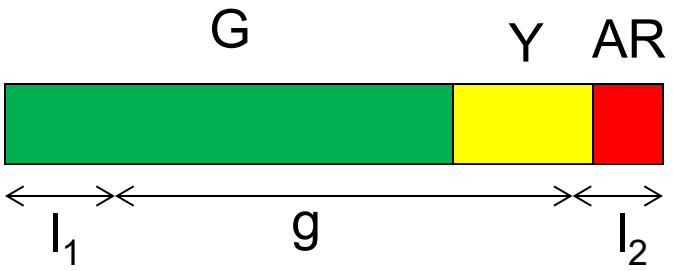
$$t_L = l_1 + l_2$$

- Where:
- t_L = total lost time per phase, s/phase
- l_1 = start-up lost time in seconds, and
- l_2 = clearance lost time in seconds.

Effective green time

- Effective green time is the time during which a traffic movement is effectively utilizing the intersection. The effective green time is calculated as

$$g = G + Y + AR - t_L$$



where:
g = effective green time for a traffic movement in seconds,
G = displayed green time for a traffic movement in seconds,
Y = displayed yellow time for a traffic movement in seconds,
AR = displayed all-red time in seconds, and
t_L = total lost time for a movement during a cycle in seconds.

$t_L = l_1 + l_2$ $\quad$ $l_2 = Y - AR - e$ $\quad$ **Finally,** $g = G - l_1 + e$

Establish Analysis Lane Groups

- HCM (6th edition) provides the following guidelines for establishing lane groups:

- An exclusive LT lane; multiple LT lanes form a single lane group.
- An exclusive RT lane; multiple RT lanes form a single lane group.
- Exclusive TH lanes (no turns from these lanes) on an approach form a single lane group.
- A shared LT/TH lane is treated as a separate lane group.
- A shared RT/TH lane is treated as a separate lane group.

Typical lane groups for analysis

Number of Lanes	Movements by Lanes	Movement Groups (MG)	Lane Groups (LG)
1	Left, thru., & right: 	MG 1: 	LG 1: 
2	Exclusive left:  Thru. & right: 	MG 1:  MG 2: 	LG 1:  LG 2: 
2	Left & thru.:  Thru. & right: 	MG 1: 	LG 1:  LG 2: 
3	Exclusive left:  Exclusive left:  Through: Through: Thru. & right: 	MG 1:  MG 2: 	LG 1:  LG 2:  LG 3: 

Capacity of a lane group

- The capacity of an intersection lane group is given by:

$$c_i = s_i\left(g_i / C\right)$$

where:

c_i = capacity of lane group i, veh/h,

s_i = saturation flow rate for lane group i, veh/h

g_i = effective green time for lane group i, s

C = signal cycle length, s

Example

- Consider a given lane group at a signalized intersection with the following known characteristics:
- Cycle length, C = 60 sec
- Green time, G = 27 sec
- Yellow plus all-red, Y = 4 sec
- Saturation headway, h = 2.4 s/veh
- Start-up lost time, l1 = 2.0 sec
- Clearance lost time, l2 = 2.0 sec

Example

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- Saturation headway, h = 2.4 s/veh
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- Clearance lost time, l2 = 2.0 sec

$$c_i = s_i\left(g_i/C\right)$$

$g = G - l_1 + e$

g = 27-2+2 = 27 seconds

C = 60 seconds

$$s = \frac{3600}{h} = 1500\ veh/h$$

$$c = s\left(\frac{g}{C}\right) = 1500 * \left(\frac{27}{60}\right) = 675\,\frac{veh}{hr}$$

Development of a traffic signal phasing and timing plan

1. Determine phasing
2. Establish analysis lane groups
3. Calculate analysis flow rates and saturation flow rates
4. Determine critical lane groups and lost times
5. Calculate cycle length
6. Allocate green time
7. Calculate yellow & clearance intervals
8. Check pedestrian crossing time

Treatment of left turns

- Two general guidelines for determining whether or not a particular left turn movement requires a protected phase.
  - $V_{LT} \geq 200$ veh/hr
  - $X_{prod} = V_{LT} \times V_0 \geq N$

  where:

  V_{LT} = left-turn flow rate, veh/h

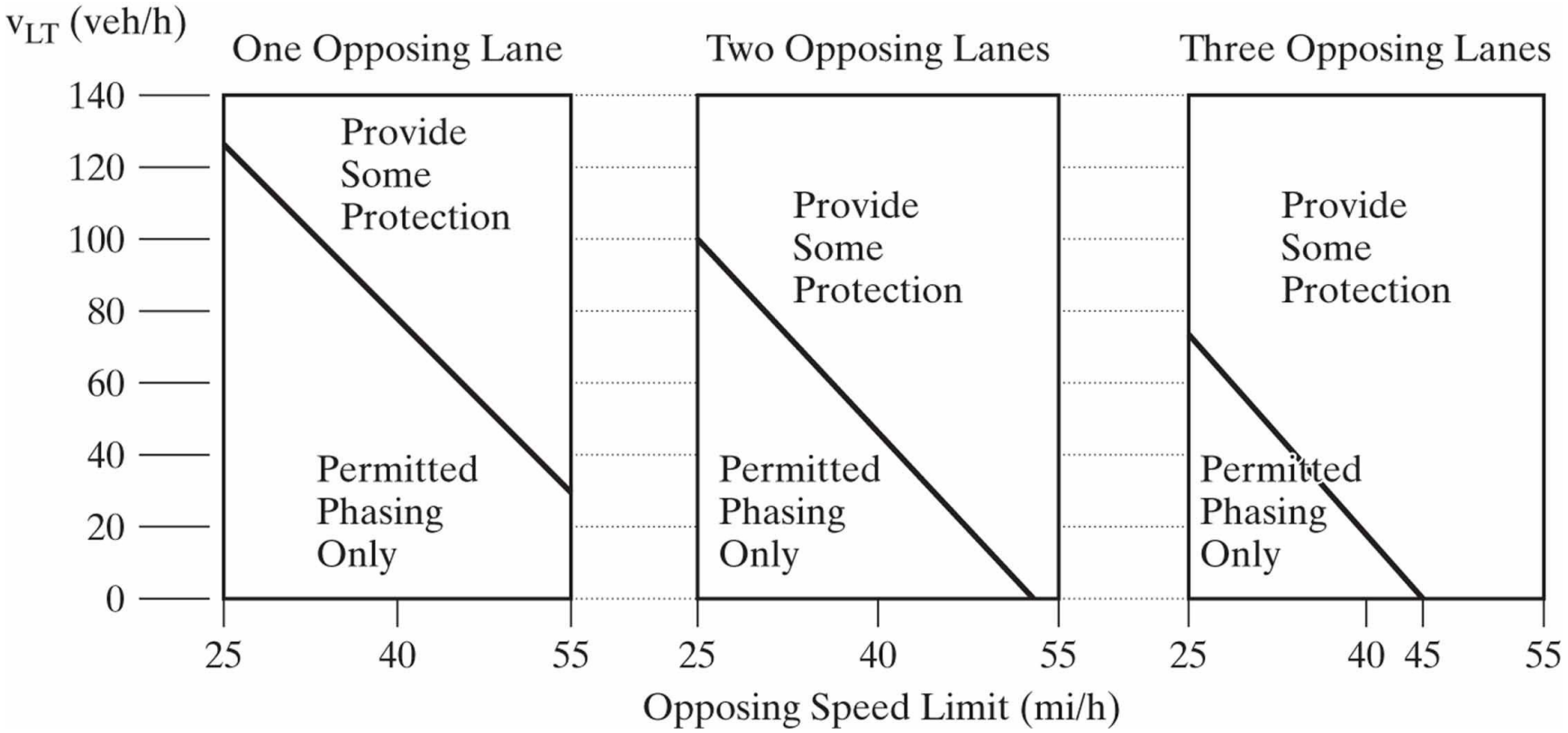
  V_0 = opposing through movement flow rate, veh/h

  N = 50,000 for one opposing lane; 90,000 for two opposing lanes; and 110,000 for three opposing lanes.

Treatment of left turns

- ITE's Traffic Engineering Handbook states that permitted phasing should be provided when the following conditions exists:
  - The left-turn demand flow within the peak hour against the speed limit for opposing traffic falls within the "permitted" portion of the exhibit, see Figure in next slide.
  - The sight distance for left-turning vehicles is not restricted.
  - Fewer than eight left-turn accidents have occurred within the last three years at any one approach with permitted-only phasing.

Figure Recommended Selection Criteria for Left-Turn Protection



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Treatment of left turns

- ITE's Traffic Engineering Handbook states fully protected phasing is recommended when any two of the following criteria are met:
  - Left-turn flow rate is greater than 320 veh/h.
  - Opposing flow rate is greater than 1,000 veh/h.
  - Opposing speed limit is greater than or equal to 45 mi/h.
  - There are two or more left-turn lanes.

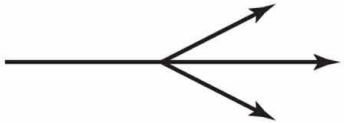
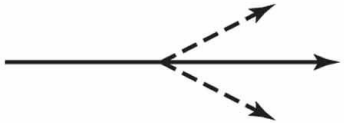
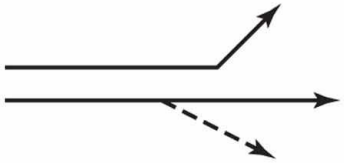
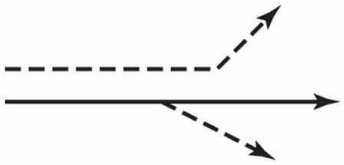
Treatment of left turns

- ITE's Traffic Engineering Handbook states fully protected phasing is also recommended when **any of the following conditions exist**:
  - There are three opposing traffic lanes, and the opposing speed is 45 mi/hr or greater.
  - Left-turn flow rate is greater than 320 and the percentage of heavy vehicles exceeds 2.5%.
  - The opposing flow rate exceeds 1,100 veh/h, and the percentage of left turns exceeds 2.5%.
  - Seven or more left-turn accidents have occurred within three years under compound phasing.
  - The average stopped delay to left-turning traffic is acceptable for fully protected phasing, and the engineer judges that additional left-turn accidents would occur under the compound phasing option.



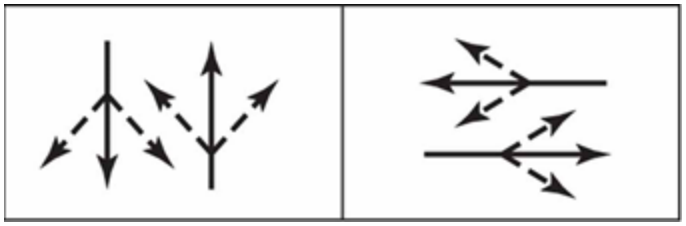
Fundamentals of Signal Timing and Design: Pre-timed Signals

Symbols used in Signal Phases Diagrams

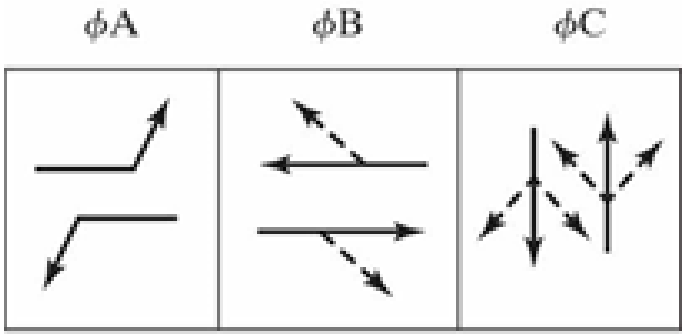
Through movement without turning movement.	
Through movement with protected right and left turns from shared lanes.	
Through movement with permitted right and left turns from shared lanes.	
Through movement with protected left turn from exclusive lane and permitted right turn from shared lane.	
Through movement with permitted left turn from exclusive lane and permitted right turn from shared lane.	

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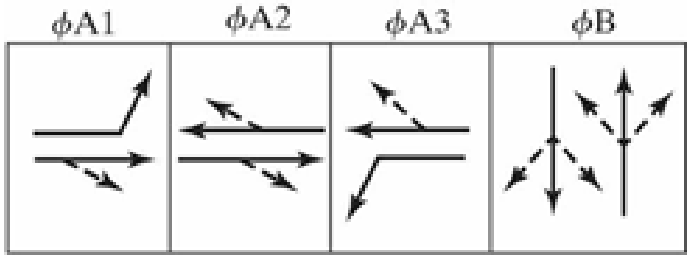
Two-Phase Signal



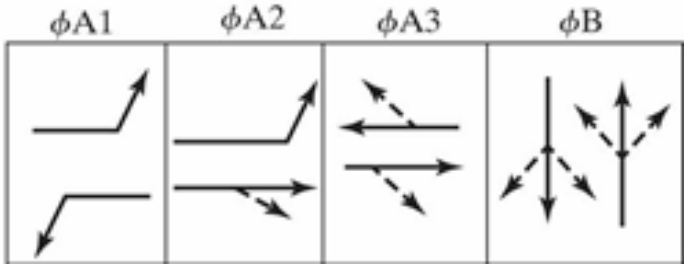
Exclusive Left-Turn Phase



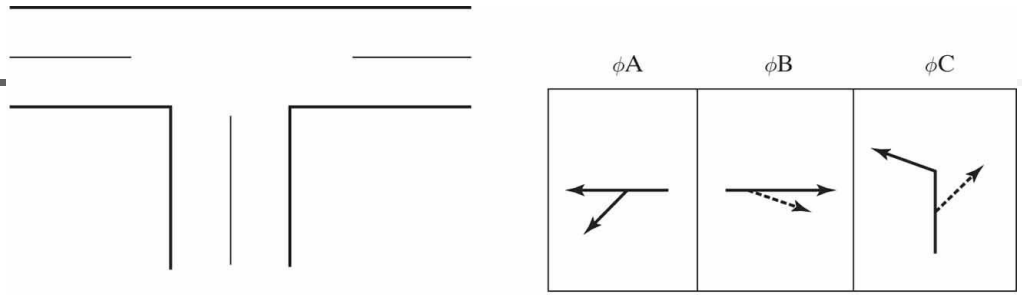
Leading and Lagging Green



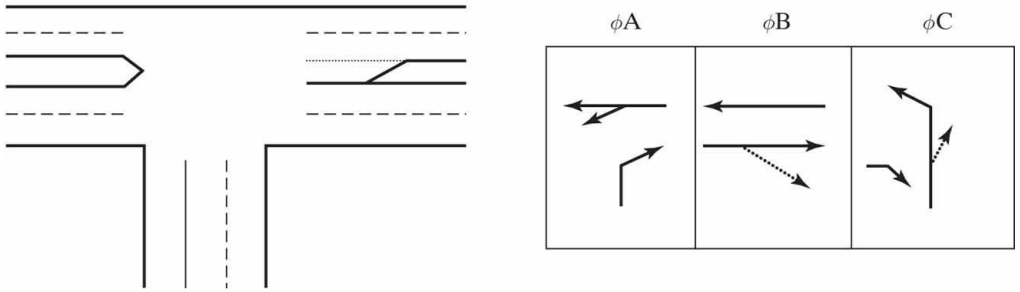
Exclusive Left-Turn Phase Plus Leading Green Phase



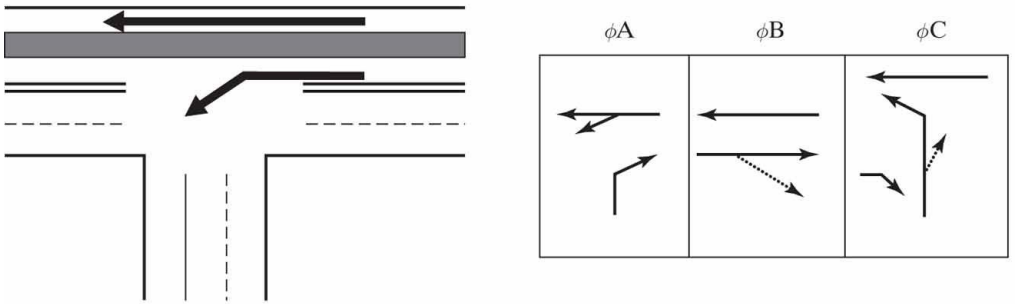
Signalization options at a T-intersection



(a) T-Intersection, No LT Lane, Protected Phasing



(b) T-Intersection, LT Lane, Protected Phasing



(c) T-Intersection, Channelized Through Movement

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General considerations in signal phasing

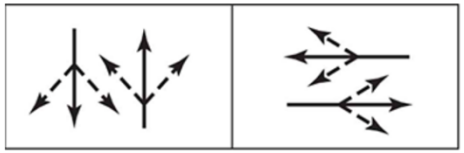
- **Additional phases generally lead to additional delay**, which must be weighed against the safety and improved efficiency of protected left turns.
- Although increasing the number of phases increases the total lost time in the cycle, the **offsetting benefit** is an increase in affected left-turn saturation flow rates.
- **All phase plans must be implemented in accordance with the standards and criteria of the MUTCD**, and they must be accompanied by the necessary signs, markings, and signal hardware needed to identify appropriate lane usage.
- The phase plan must be **consistent with the intersection geometry, lane-use assignments, volumes and speeds, and pedestrian crossing requirements.**

Development of a traffic signal phasing and timing plan

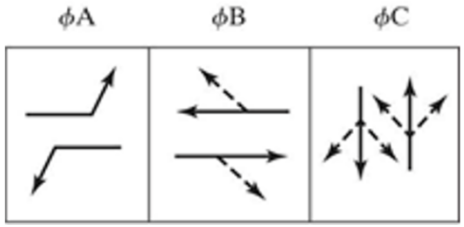
1. Determine phasing
2. Establish analysis lane groups
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6. Allocate green time
7. Calculate yellow & clearance intervals
8. Check pedestrian crossing time

1. Determine phasing

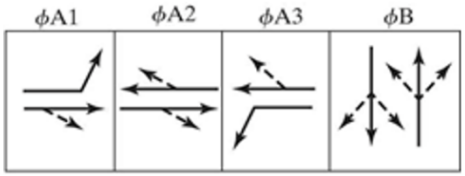
Two-Phase Signal



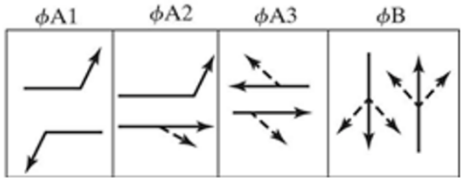
Exclusive Left-Turn Phase

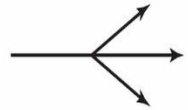
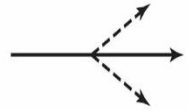
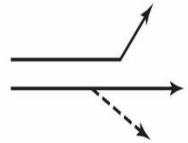
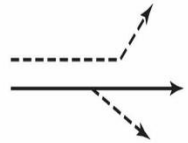


Leading and Lagging Green



Exclusive Left-Turn Phase Plus Leading Green Phase



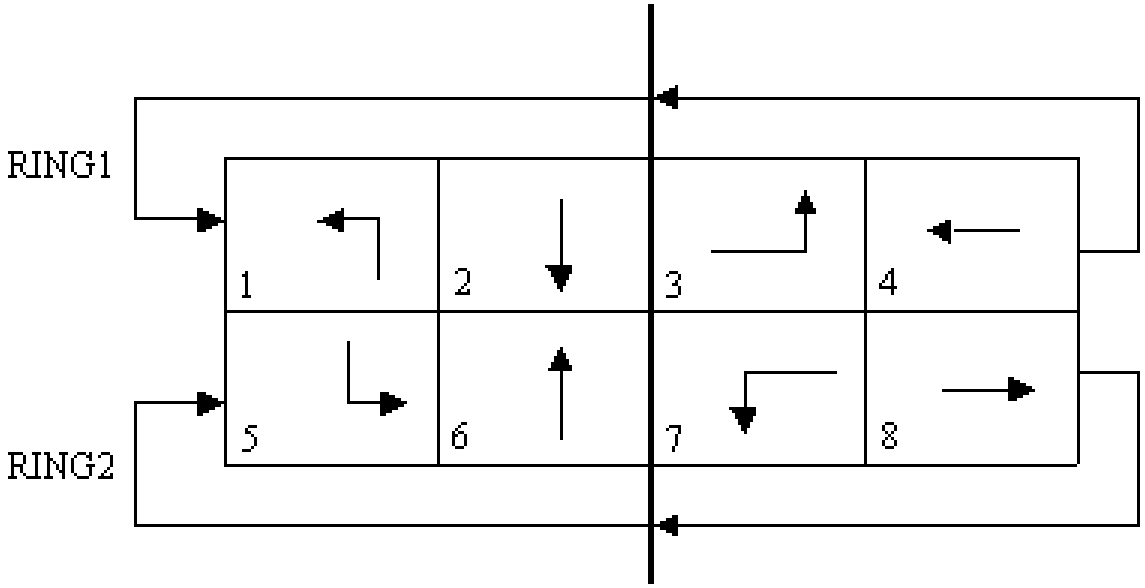
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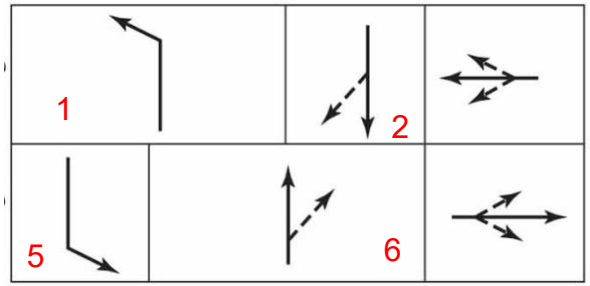
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1. Determine phasing

- Phase diagram vs. ring diagram
  - **Phase diagram**: shows all movements being made in a given phase within a single block of the diagram.
  - **Ring diagram:** shows which movements are **controlled by which ring on a signal controller.** A ring of a controller generally controls traffic from one approach or lane group.

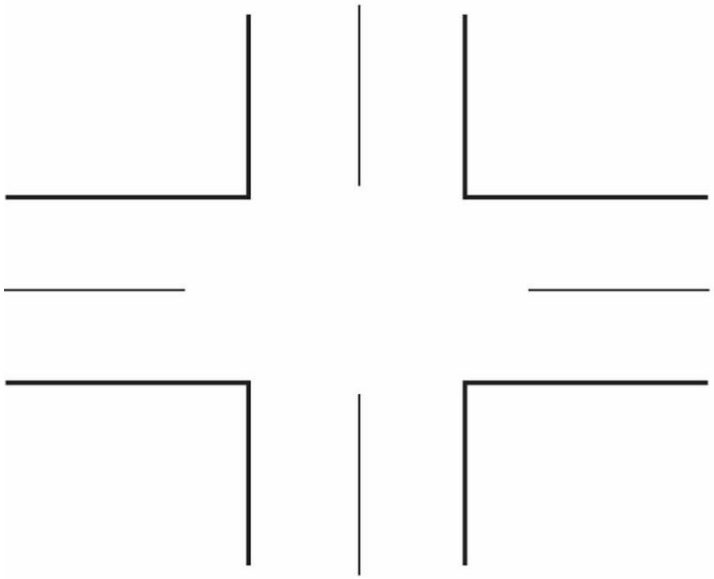
Dual-ring configuration



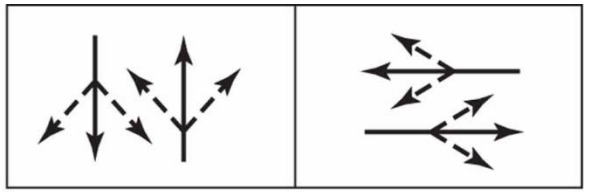


Phase diagram vs. ring diagram example 1

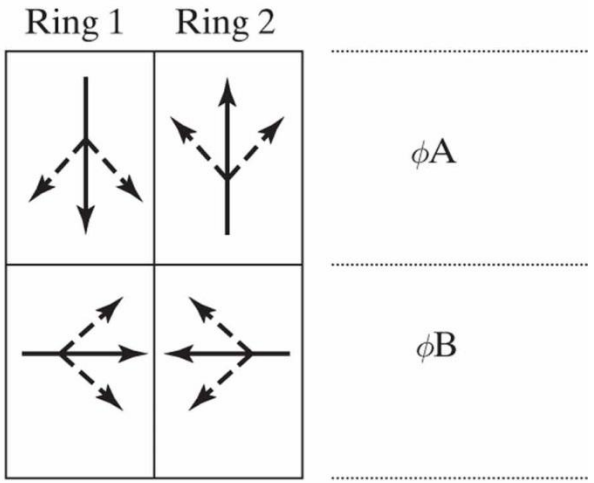
- While two opposing through movements are shown in one block of a phase diagram, each movement is shown separately in a ring diagram.



(a) Intersection Layout (exclusive LT/RT lanes optional)



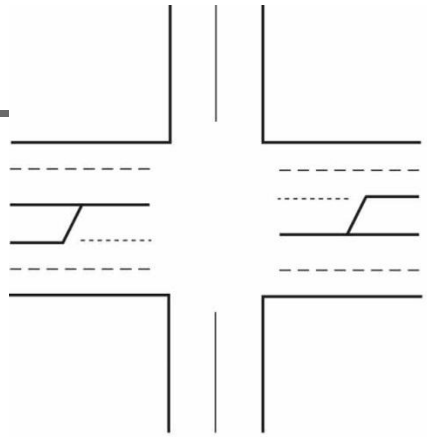
(b) Phase Diagram



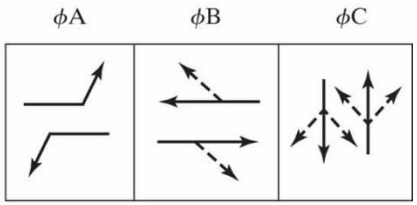
(c) Ring Diagram

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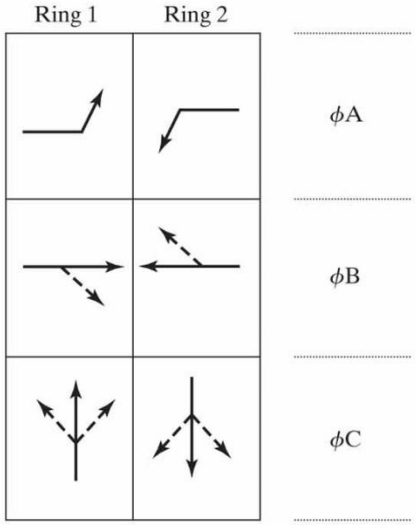
Phase diagram vs. ring diagram example 2



(a) Intersection Layout



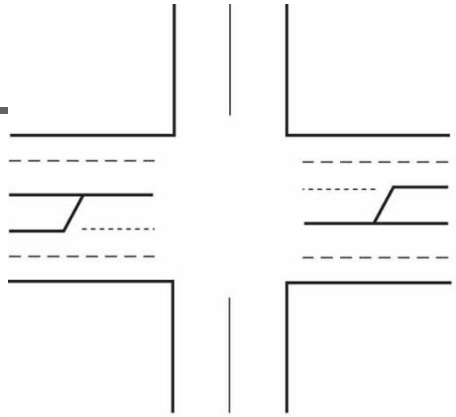
(b) Phase Diagram



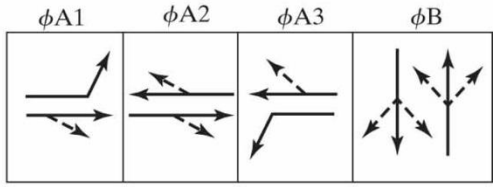
(c) Ring Diagram

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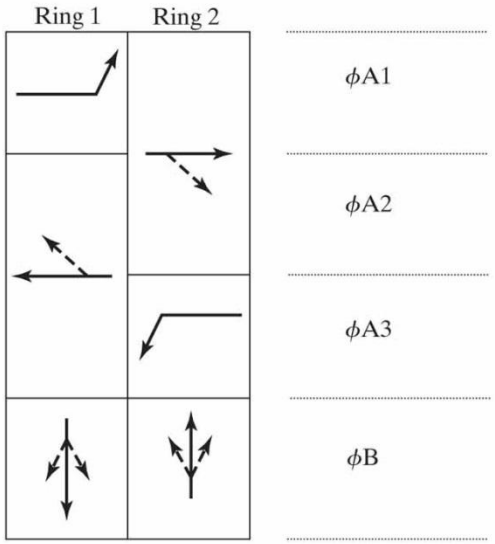
Phase diagram vs. ring diagram example 3



(a) Intersection Layout



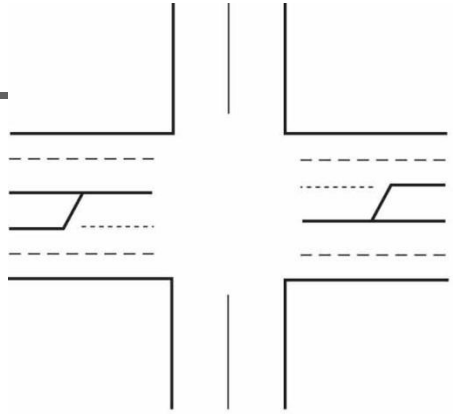
(b) Phase Diagram



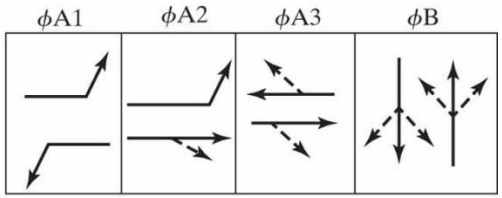
(c) Ring Diagram

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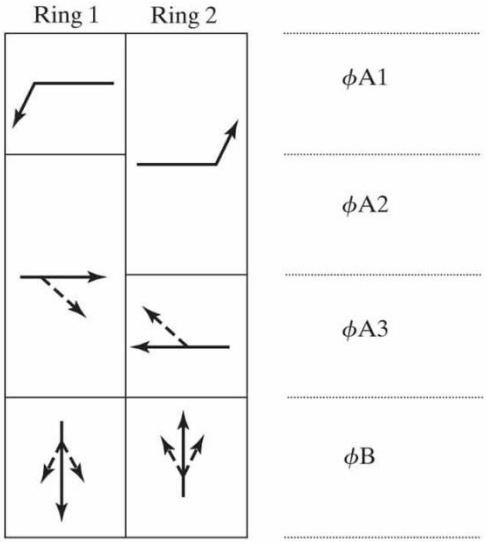
Phase diagram vs. ring diagram example 4



(a) Intersection Layout



(b) Phase Diagram



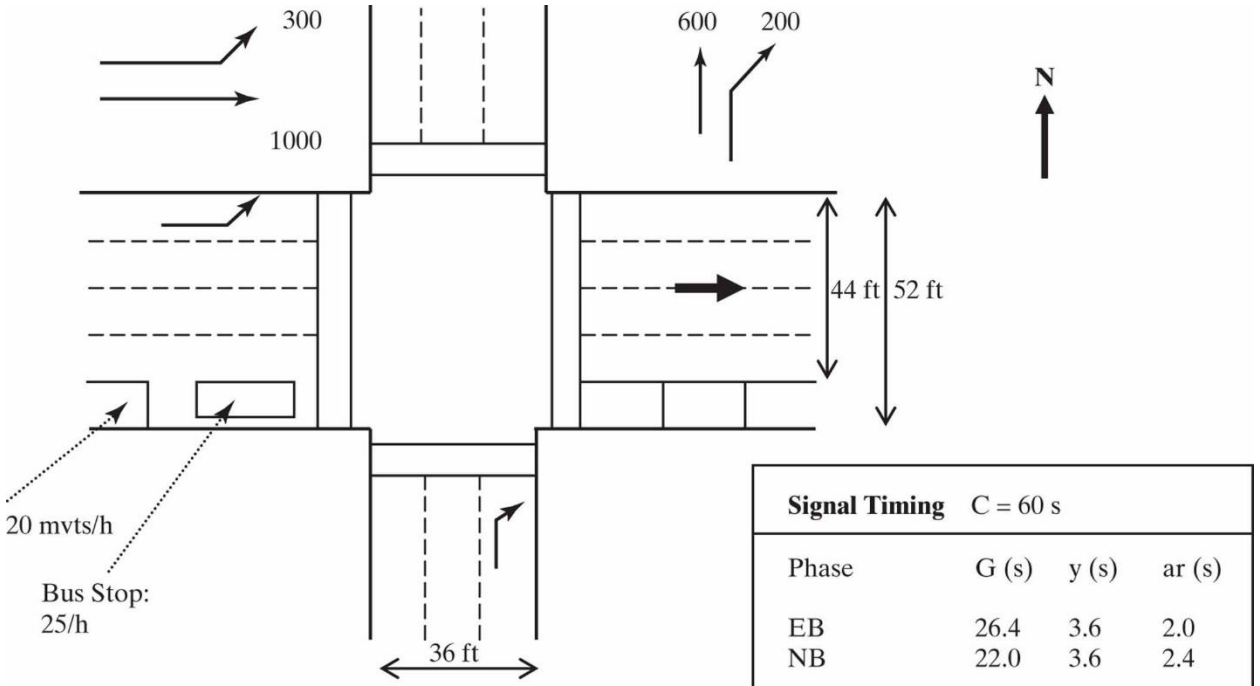
(c) Ring Diagram

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2. Establish analysis lane groups

- HCM (6th edition) provides the following guidelines for establishing **lane groups**:
- An exclusive LT lane; multiple LT lanes form a single lane group.
- An exclusive RT lane; multiple RT lanes form a single lane group.
- Exclusive TH lanes (no turns from these lanes) on an approach form a single lane group.
- A shared LT/TH lane is treated as a separate lane group.
- A shared RT/TH lane is treated as a separate lane group.

3. Calculate analysis flow rates and saturation flow rates



Signal Timing	C = 60 s		
Phase	G (s)	y (s)	ar (s)
EB	26.4	3.6	2.0
NB	22.0	3.6	2.4

HV% = 10% in all mvts.
100 peds/h in all crosswalks.
Crosswalk widths = 10 ft
Grades: NB +3%; EB level.

PHF = 0.90.
No RTOR.
Arrival Types: NB = 3; EB = 5.
Area Type: Fringe Area

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Calculation for flow rate for example problem

Table 24.8: Flow Rate Computations for Sample Problem

Movement	*V* (veh/h)	*PHF*	Flow Rate (veh/h)	Lane Group Flow Rate (veh/h)
EB-LT	300	0.90	333	333
EB-TH	1000	0.90	1111	1111
NB-TH	600	0.90	667	667
NB-RT	200	0.90	222	222

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Estimation of saturation flow rates for example problem

DETERMINING SATURATION FLOW RATE

A saturation flow rate for each lane group is computed according to Equation 16-4. The saturation flow rate is the flow in vehicles per hour that can be accommodated by the lane group assuming that the green phase were displayed 100 percent of the time (i.e., g/C = 1.0).

$$s = s_o N f_w f_{HV} f_g f_p f_{bb} f_a f_{LU} f_{LT} f_{RT} f_{Lpb} f_{Rpb} \quad (16\text{-}4)$$

where

s	=	saturation flow rate for subject lane group, expressed as a total for all lanes in lane group (veh/h);
s_o	=	base saturation flow rate per lane (pc/h/ln);
N	=	number of lanes in lane group;
f_w	=	adjustment factor for lane width;
f_{HV}	=	adjustment factor for heavy vehicles in traffic stream;
f_g	=	adjustment factor for approach grade;
f_p	=	adjustment factor for existence of a parking lane and parking activity adjacent to lane group;
f_{bb}	=	adjustment factor for blocking effect of local buses that stop within intersection area;
f_a	=	adjustment factor for area type;
f_{LU}	=	adjustment factor for lane utilization;
f_{LT}	=	adjustment factor for left turns in lane group;
f_{RT}	=	adjustment factor for right turns in lane group;
f_{Lpb}	=	pedestrian adjustment factor for left-turn movements; and
f_{Rpb}	=	pedestrian-bicycle adjustment factor for right-turn movements.

Estimation of saturation flow rates for sample problem

Table 24.11: Estimation of Saturation Flow Rates Using Equation 24-11

Lane Group	s_o (pc/hg/ln)	N (lanes)	f_w	f_{HV}	f_g	f_p	f_{bb}	f_a	f_{LU}	f_{RT}	f_{LT}	f_{Rpb}	f_{Lpb}	s (veh/hg)
EB LT	1900	1	1.000	0.909	1.000	1.000	1.000	1.000	1.000	1.000	0.952	1.000	0.932	**1533**
EB TH	1900	3	1.000	0.909	1.000	0.933	0.967	1.000	0.908	1.000	1.000	1.000	1.000	**4245**
NB TH	1900	2	1.000	0.909	0.985	1.000	1.000	1.000	0.952	1.000	1.000	1.000	1.000	**3240**
NB RT	1900	1	1.000	0.909	0.985	1.000	1.000	1.000	1.000	0.850	1.000	0.918	1.000	**1328**

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v/s ratios

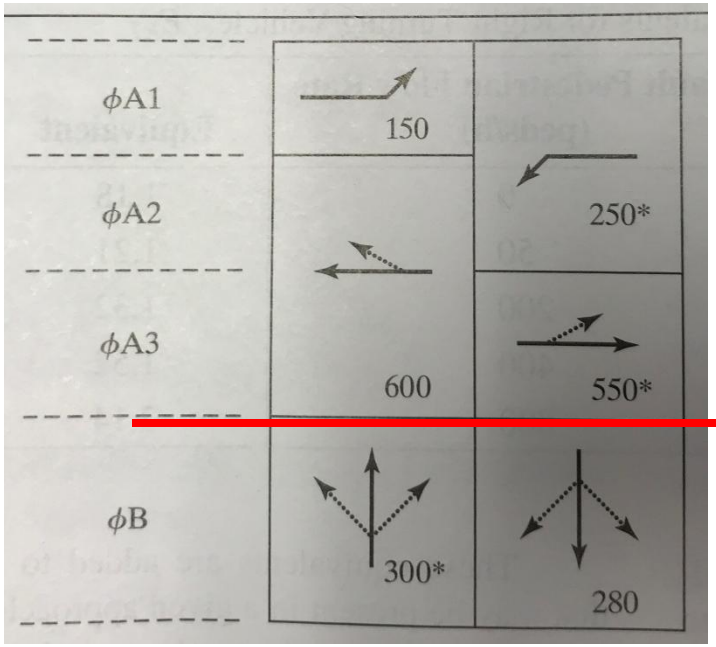
Lane Group	v (veh/h)	s (veh/h)	v/s
EB LT	333	1533	0.217
EB TH	1111	4245	0.262
NB TH	667	3240	0.206
NB RT	222	1328	0.167

4. Determine critical lane groups and lost times

- To estimate an appropriate cycle length and to split the cycle into appropriate green times for each phase, it is necessary to find the **critical-lane volume for each discrete phase** or portion of the cycle.
- **Critical lane volume**: the per-lane volume that controls the required length of a particular phase.
- *Traffic demand is not evenly distributed among lanes at intersection, one of these lanes will have the most intense traffic demand and the signal must be timed to accommodate traffic in this lane, i.e. critical lane.*

Determining the sum of critical-lane volume

- Find sum of critical-lane volume
  - When phasing involves overlapping elements, then ring diagrams must be carefully examined to determine which flows constitute CLVs.
  - Add volume for each ring before barrier.



150+600=750 tvu
250+550=800 tvu

280 tvu
300 tvu

Critical-lane volume
Vc = 800+300 = 1100 tvu

Determining lost time for each phase and cycle

- Total lost time = start-up lost time + clearance lost time = l_1+l_2
- HCM recommends the following default values.
  - Start-up lost time, l_1 = 2 sec/phase
  - Motorist use of yellow and all-red, e = 2 sec/phase.
- $t_{Li} = l_1 + l_2$ and $l_2 = Y + AR - e$, for each phase i
- Total lost time for cycle $L = \sum_{i}^{n} t_{Li}$

Yellow Interval - Y

- The ITE's method for determining yellow or change interval:

$$Y = t + \frac{1.47 S_{85}}{2a + (64.4 \times 0.01 G)}$$

Y	=	yellow time, s
t	=	driver perception/reaction time, s (typically, 1 sec is used)
S_{85}	=	85th percentile speed of approaching traffic, mi/hr
a	=	deceleration rate of vehicles, ft/s^2 (typically, 10 ft/s^2 is used).
G	=	grade of approach, %.

All-red interval - AR

- ITE's method for determining all-red:
  - When there is no/little pedestrian:

$$AR = \frac{w + L}{1.47 S_{15}}$$

  - When there is significant pedestrian crossing:

$$AR = \frac{P + L}{1.47 S_{15}}$$

AR = all-red time, s
w = distance from stop line to the far side of the farthest conflicting traffic lane, ft
P = distance from the stop line to the far side of the farthest conflicting crosswalk, ft
L = length of a standard vehicle, usually taken as 18 to 20 ft
S_{15} = 15th percentile speed of approaching traffic, or speed limit as appropriate, ft/s.

Estimating S_{15} and S_{85}

- If only the average approach speed is known, the percentile speeds may be estimated as follows.
  - $S_{15} = S - 5$
  - $S_{85} = S + 5$
  - Where S is the average speed, mi/hr

Example

- Compute the appropriate change and clearance intervals for a signalized intersection approach with the following characteristics:
  - Average approach speed = 35 mi/hr
  - Grade = –2.5%
  - Distance from STOP line to far side of the most distant lane = 48 ft
  - Distance from STOP line to far side of the most distant crosswalk = 60 ft
  - Standard vehicle length = 20 ft
  - Reaction time = 1 sec
  - Deceleration rate = 10 ft/s^2
  - Little pedestrians present

Example

- Compute the appropriate change and clearance intervals for a signalized intersection approach with the following characteristics:
  - Average approach speed = 35 mi/hr
  - Grade = –2.5%
  - Distance from STOP line to far side of the most distant lane = 48 ft
  - Distance from STOP line to far side of the most distant crosswalk = 60 ft
  - Standard vehicle length = 20 ft
  - Reaction time = 1 sec
  - Deceleration rate = 10 ft/s^2
  - Little pedestrians present

$$Y = t + \frac{1.47S_{85}}{2a + (64.4 \times 0.01G)}$$

$$AR = \frac{w + L}{1.47S_{15}}$$

$$Y = 1 + \frac{1.47 * (35 + 5)}{2 * 10 + (64.4 * -0.025)} = 4.2\ s$$

$$AR = \frac{48 + 20}{1.47 * (35 - 5)} = 1.5\ s$$

5. Calculate Desired Cycle Length

- The desired cycle length is computed as follows.

$$C_{des} = \frac{L}{1 - \frac{V_c}{1700 * PHF * (v/c)}}$$

C = cycle length in seconds (typically rounded up to the nearest 5-second increment in practice),
L = total lost time for cycle in seconds,
v/c = target v/c ratio (typically between 0.8 and 0.95),
PHF = peak hour factor.

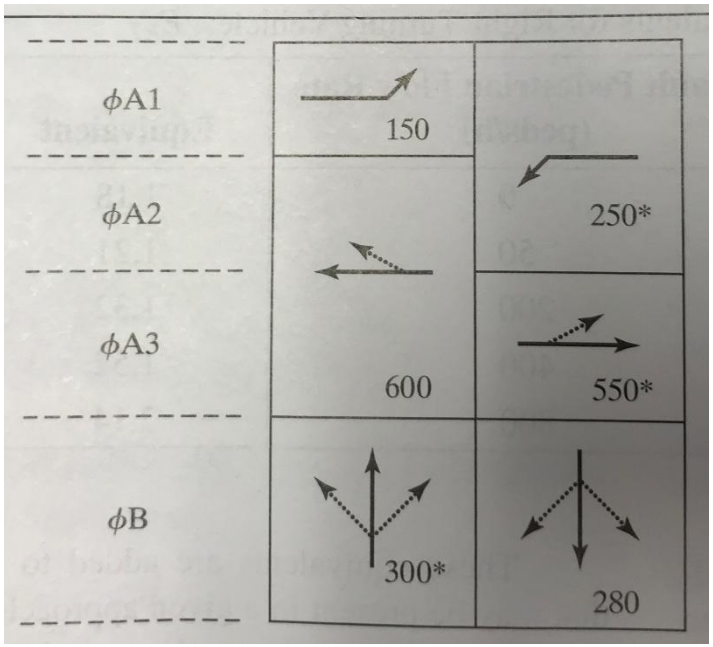
5. Calculate Desired Cycle Length, Webster's Formula

- The desired cycle length is computed as follows.

$$C = \frac{L * X_c}{X_c - V_c}$$

C = cycle length in seconds (typically rounded up to the nearest 5-second increment in practice),
L = total lost time for cycle in seconds,
V_c = critical lane v/c ratio,
X_c = peak hour factor.

5. Calculate Desired Cycle Length



Critical-lane volume
Vc = 800+300 = 1100 tvu
PHF = 0.92
Target v/c = 0.9
Total lost time L = 12 s

$$C_{des} = \frac{L}{1 - \frac{V_c}{1700 * PHF * (v/c)}}$$

$$= \frac{12}{1 - [\frac{1100}{1700 * 0.92 * 0.9}]} = 54.8 \cong 55\ s$$

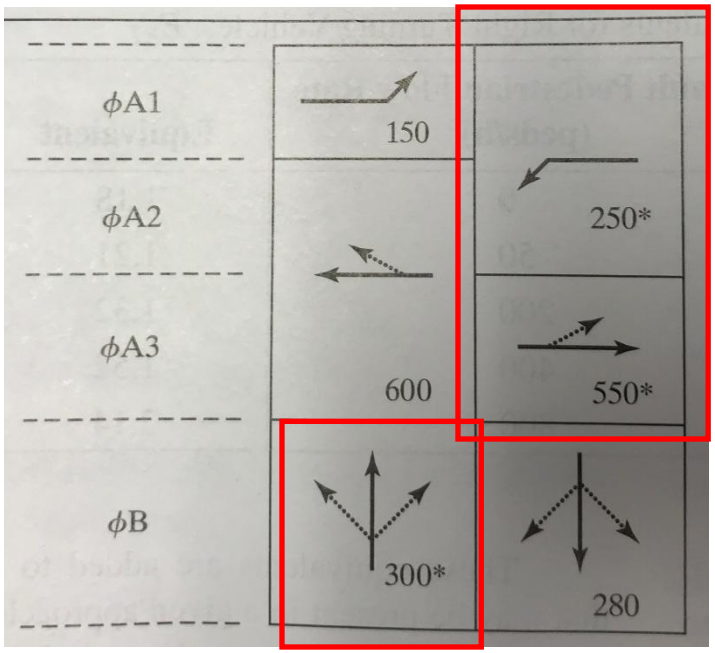
6. Allocate green time

- Effective green time is distributed in proportion to the critical flow to saturation ratios for each phase or subphase, as follows.

$$g_i = g_{TOT} * (\frac{v_{ci}}{v_c})$$

g_i	=	effective green time for phase *i*,
v_{ci}	=	flow for critical lane group *i*,
g_{TOT}	=	total effective green time per cycle. g_{TOT} = C – L
V_c	=	sum of flow for critical lane groups

6. Allocate green time



Critical-lane volume
Vc = 800+300 = 1100 tvu
Total lost time L = 12 s
Desired Cycle time C = 55 s

g_TOT = 55-12 = 43 s

$$g_{A1+A_2} = 43 * \left(\frac{250}{1100}\right) = 9.8\ s$$

$$g_{A3} = 43 * \left(\frac{550}{1100}\right) = 21.5\ s$$

Time for A1,A2,A3 = 9.8+21.5=31.3

$$g_{A1} = 31.3 * \left(\frac{150}{150+600}\right) = 6.3\ s$$

$$g_{B} = 43 * \left(\frac{300}{1100}\right) = 11.7\ s$$

7. Calculate yellow & clearance intervals

- Already shown in step 4, Y and AR

8. Pedestrian crossing time

- The minimum pedestrian green time is given by

$$G_p = 3.2 + \left(2.7 \times \frac{N_{ped}}{W_E}\right) + \left(\frac{L}{S_p}\right) \text{ for } W_E > 10 \text{ ft}$$

$$G_p = 3.2 + \left(2.7 \times N_{ped}\right) + \left(\frac{L}{S_p}\right) \text{ for } W_E \leq 10 \text{ ft}$$

G_p	=	minimum pedestrian green time, s
3.2	=	pedestrian start-up time
L	=	crosswalk length, ft,
S_p	=	walking speed of pedestrians, usually taken as 3.5 ft/s
N_{ped}	=	number of pedestrians crossing during an interval
W_E	=	effective crosswalk width, ft

Pedestrian crossing time

- The WALK interval may be longer than the minimum required by pedestrians if the vehicular green is longer than needed.
- The total length of the WALK + Flashing DON'T WALK intervals can be considered in a number of ways:
  - $G_p \leq G + Y + AR$
  - $G_p \leq G + Y$
  - $G_p \leq G$
- When the minimum pedestrian green time is not met, the signal should be retimed to provide the minimum pedestrians need in all cycles. This must be done in a manner that also maintains the vehicular balance of green times and results in a longer cycle length.

Example

Table 19.7: A Sample Signal Timing

Phase	Green Time, G (s)	Yellow Time, y (s)	All-Red Time, ar (s)	Lost Time, t_L (s)	Minimum Pedestrian Green, G_p (s)
A	40	3.0	2.0	5.0	20.0
B	15	3.0	2.0	5.0	30.0